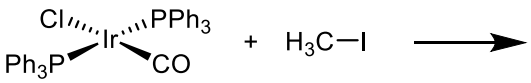
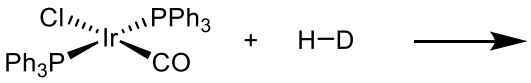
Inorganic Chemistry
Cumulative Exam
2-16-24

Inorganic Chemistry in Homogeneous Solution

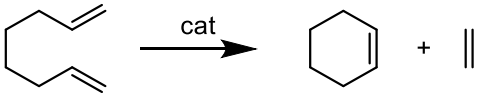
Answer any five of the following ten questions (20 points each). Only your first five answers will be graded.

1. $[Co(NH_3)_4Cl_2]Cl$ can be made in two forms one is purple the other yellow. Explain the source of this difference.
2. Some transition metal complexes show intense charge transfer absorption bands that shift significantly (red to green) with the choice of solvent in which the spectra is obtained. Provide a reasonable hypothesis for this observation. The answer you provide will be judged on the scientific soundness of your hypothesis and not how well it fits to current models. Hint: The fact that the band is due to charge transfer is very important.
3. Molybdenum hexacarbonyl has an A1 CO stretching frequency is 2001 cm^{-1} Tetrasodium tetracarbonylmolybdate has an A1 stretching frequency of 1463 cm^{-1} Why?
4. Predict the products of the following two reactions and provide a complete mechanism for each reaction.





5. Provide a catalyst and complete mechanism for the following reaction.



Solid State Inorganic Chemistry

6. Unlike classic coordination compounds the absorption spectra of quantum dots are often discussed but rarely in terms of the chemical composition. Why?
7. Emeralds are green yet one could write the chemical formula as $Be_3Al_2(SiO_3)_6$ which is the same as the mineral beryl which is colorless. The same can be said for the form of aluminum oxide called rubies. Why?
8. The crystal structures of many salts can be rationalized in terms of a cubic close packing or hexagonal close packing model. Describe these two models.
9. Many metals have moderately low melting points but very high boiling points. For example, aluminum melts at 660 ^{o}C and boils at 2467 ^{o}C whereas bromine, a nonmetal, melts at -8 ^{o}C and boils at 58 ^{o}C. Rationalize this trend using any bonding theory.
10. Superconductivity was first discovered in 1911. In the 1950's Bardeen, Cooper and Schrieffer described a theory explaining superconductivity. A) What experimental observations would lead you to conclude that your material is a superconductor? B) Describe BCS theory.